

DATA ANALYTICS

SECTION 1: Core Concepts

Q1. Definitions with example:

Data

Data refers to raw facts or values collected from activities. On its own, data does not provide meaning.

Example: Daily sales numbers recorded in a store.

Data Analytics

Data analytics is the process of examining data to identify patterns, trends, and insights that help in decision-making.

Example: Analyzing sales data to find the most profitable product.

Difference between Data and Information

Data is unprocessed raw facts, while information is processed data that has meaning.

Example: Individual student marks are data; the class average is information.

Q2. Why companies invest in data analytics?

Companies invest in data analytics to:

1. Make better decisions based on facts rather than assumptions.
2. Understand customer behavior and preferences.
3. Reduce costs by identifying inefficiencies.
4. Increase revenue through demand forecasting and personalization.
5. Reduce risks by detecting issues early.

SECTION 2: Types of Data

Q3. Data classification:

Scenario	Type of Data	Nature
Customer age	Structured	Quantitative
Netflix movie review text	Unstructured	Qualitative
Bank transaction table	Structured	Quantitative
Uber GPS coordinates	Semi-Structured	Quantitative
Amazon product ratings (1–5)	Structured	Quantitative

Q4. Structured form of feedback:

Feedback_text	Sentiment	Issue_type
The delivery was late and customer support was poor	Negative	Delivery & Support

SECTION 3: Analytics Types Mapping

Business Question	Analytics Type	Reason
Why are users uninstalling our app?	Diagnostic Analytics	Finds reasons behind a problem
How many users logged in yesterday?	Descriptive Analytics	Summarizes past data
What will next month's revenue be?	Predictive Analytics	Forecasts future outcomes
Which action should reduce churn?	Prescriptive Analytics	Recommends best actions

SECTION 4: Industry Use Case – Amazon

Q6. Data collected & objectives:

- Customer purchase history – understand buying behavior.
- Product ratings & reviews – improve product quality.
- Search history – personalize recommendations.
- Delivery data – optimize logistics.
- Payment data – prevent fraud.

Q7. Analytics use cases:

- Descriptive: Daily sales report – understand what happened.
- Diagnostic: Drop in sales analysis – find reasons.
- Predictive: Demand forecasting – predict future sales.
- Prescriptive: Discount strategy – recommend best action.

SECTION 5: Analyst Thinking Test

Situation: Company revenue is falling.

Questions to ask first:

- Which product or region is affected?
- Is customer demand decreasing?
- Are prices or competitors impacting sales?

Data required:

- Sales data
- Customer data
- Marketing performance data
- Cost and pricing data

Metrics to analyze:

- Revenue trends
- Profit margin
- Customer churn rate
- Conversion rate

SECTION 6: Executive Summary for CEO

What is Data Analytics?

Data analytics is the structured use of business data to generate meaningful insights.

It transforms raw information into clear and actionable understanding.

Analytics helps explain past performance and current business conditions.

It identifies patterns, trends, and areas requiring attention.

Overall, it enables informed understanding across the organization.

Why Data Analytics Matters to the Business?

Data analytics supports fact-based decision-making at the leadership level.

It minimizes risk by reducing reliance on assumptions and intuition.

Analytics improves operational efficiency and resource allocation.

It enhances understanding of customers, markets, and performance drivers.

Ultimately, it enables sustainable growth and competitive advantage.